How are mass extinctions shaping biodiversity?

Confirmation Review

Caleb Scutt
Supervisors: Thomas Guillerme (UoS), Gavin Thomas (UoS) & Natalie Cooper (NHM)
Advisor: Karl Evans (UoS)

July 2025

Contents

I Abstract **2**

II Background Review **2**

1 Morphology, Disparity and Extinction **2**
1.1 Mass extinctions . . . 2
1.2 Rebound dynamics . . . 3
1.3 Disparity . . . 3
1.4 Extinction selectivity . . . 4
1.5 Disparity metric . . . 4

2 Eutherian disparity across K-Pg Mass Extinction **5**
2.1 Suppression hypothesis . . . 5
2.2 Early rise hypothesis . . . 5
2.3 Delayed rise hypothesis . . . 6

3 Disparity Analysis Methods **8**

4 Phylogenetic hypotheses **9**

5 Ancestral state estimation **9**
5.1 Application of ancestral state estimation . . . 10
5.1.1 Likelihood methods . . . 10
5.1.2 Continuous characters . . . 11
5.1.3 Discrete characters . . . 11
5.2 Challenges in ancestral state estimation . . . 12
5.2.1 Uncertainty in estimates . . . 12
5.2.2 Sensitivity to fossil sampling . . . 13
5.2.3 Model dependency . . . 13

6 Statistical tests in disparity **14**

III Thesis plan **15**

7 Aims and objectives **15**

8 Chapter One: Eutherian phylogenies **16**

9 Chapter Two: What is the effect of ancestral state estimations on disparity? **16**

10 Chapter Three: Measuring changes in trends in disparity through time **17**

11 Chapter Four: Eutherian disparity over K-Pg mass extinction **19**

IV Summary of research achievements **19**

12 Methods **20**
12.1 Simulations 20
12.2 Ancestral state estimation 20
12.2.1 Discrete characters 21
12.2.2 Continuous characters 21
12.3 Measuring raw distance 21
12.4 Measuring error in disparity 22
12.5 Extinction selectivity 22

13 Results and Discussion **22**
13.1 Discrete characters 22
13.2 Continuous characters 25

14 Future directions **27**

Part I

Abstract

Part II

Background Review

1 Morphology, Disparity and Extinction

1.1 Mass extinctions

Extinction is a fundamental evolutionary process that, alongside speciation, continuously shapes patterns of biodiversity through deep time; over 99% of all species that have ever lived are now extinct (Raup, 1981). However, extinction rates are not constant and occur in pulses across background taxonomic turnover (Raup and Sepkoski, 1982). *Mass extinctions* are specific events that wipe out over 75% of species and have happened five times over the last 500 million years (Raup and Sepkoski, 1982). On a macroevolutionary scale, mass extinctions mark the most dramatic species composition transitions in Earth history (Jablonski, 2001). Entire branches are trimmed from the tree of life, while the evolutionary trajectories of survivors are altered in the wake of a transformed global biota (Hull, 2015).

1.2 Rebound dynamics

The aftermath of this drastic species turnover is typically described as 'recovery', yet Jablonski and Edie (2025) describe this as a misnomer, since this implies a return to the prior pre-extinction state. Instead, biodiversity rebounds are multifaceted and the response from the survivors of a mass extinction event may take several forms, transforming the post-extinction biota (Hull, 2015). For example, trends observed in different clades pre-mass extinction may continue unperturbed, including escalations (Hansen et al., 1999) and declines (Raup, 1981). Alternatively, the aftermath of a mass extinction can trigger diversification as surviving clades are offered ecological opportunity, filling the vacant niches left by extinct lineages.

Rebound dynamics can differ by the metric of diversity considered. A discordance between taxonomic and morphological diversity, which can be considered a proxy of ecological diversity, is common in empirical macroevolutionary studies. Intuitively, a diversifying clade would exhibit simultaneous taxonomic and phenotypic diversification, which are expected to follow a pattern of early high rates that decrease through time as niches become saturated (Schluter, 2000). However, empirical evidence often finds that peak taxonomic and morphological diversity are reached at different points in a clade's history (Foote, 1993; Hopkins, 2013; Ruta et al., 2013; Wan et al., 2021). This discordance has been observed across mass extinction events, such as in graptolites across the end-Ordovician mass extinction (Bapst et al., 2012), or therapsids across the Permian-Triassic mass extinction (Ruta et al., 2013). It has been suggested that functional diversity may be density-independent across mass extinction events (Edie et al., 2018). Therefore, understanding the rebound dynamics of clades based solely on taxonomic diversity metrics may not encapsulate the full complexity of rebound dynamics (Puttick et al., 2020).

1.3 Disparity

This phenotypic diversity, often called *disparity*, has been widely used to investigate patterns of evolution in extant and extinct clades (Wills et al., 1994). Disparity refers to the extent of morphological diversity among taxa, typically quantified as the distribution or spread of those taxa within a morphospace — a multidimensional mathematical space configured by morphological similarity (where similar looking species are positioned closer together) (Guillerme, Cooper, Brusatte, Davis, Jackson, Gerber, Goswami, Healy, Hopkins, Jones et al., 2020). Changes in morphospace through time are often closely tied to shifts in ecological diversity (Lowery and Fraass, 2019; Budd, 2021). For example, expansion in morphospace may indicate a clade expanding its niche breadth during an adaptive radiation (Foote, 1994; Raup, 1994; Sidlauskas, 2008). Alternatively, a clade may move position within the morphospace, reflecting shifts in ecological niche (Hopkins, 2016).

1.4 Extinction selectivity

Mass extinctions pose numerous scenarios of disparity (Figure 1). In adaptive scenarios, disparity may increase in clades that diversify during the period of ecological opportunity following a mass extinction due to morphospace expansion. Extinction can also be random or selective based on specific traits, or regions of the morphospace, that go extinct. Korn et al. (2013) proposed several patterns to describe different selection processes: lateral (asymmetric selection that trims one side of the morphospace), marginal (symmetric selection that removes peripheral taxa) and non-selective ('field of bullets' random process or independent of the variables defining the morphospace). Work on contemporary extinctions has also investigated extinction selectivity, with certain trait combinations making extant taxa more susceptible to extinction (Ali et al., 2023; Matthews et al., 2024).

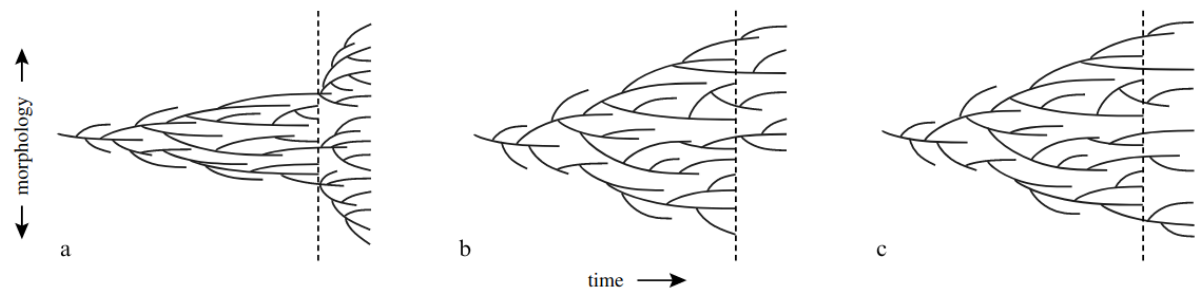

Figure 1: Three scenarios that may occur within a clade across a mass extinction (denoted by dashed line). a) adaptive radiation. b) selective extinction. c) non-selective extinction. Adapted from Grossnickle and Newham (2016).

1.5 Disparity metric

These different types of extinction events can be distinguished from each other by comparing changes in different disparity metrics. For example, a variance-based metric such as sum of variances may remain relatively unchanged during a random extinction, since no systematic reduction in trait variance has occurred (Foote, 1997; Korn et al., 2013). In contrast, a reduction in sum of variances will likely be apparent in a marginal selective extinction, where the pruning of peripheral taxa has led to a contraction in variance. Puttick et al. (2020) highlight the flaws in using a single disparity metric for assessing morphospace changes, since changes can occur in size, position and density (Guillerme, Puttick, Marcy and Weisbecker, 2020). Studies that have implemented multiple metrics have been able to identify nuanced changes in morphospace (Cole and Hopkins, 2021).

2 Eutherian disparity across K-Pg Mass Extinction

2.1 Suppression hypothesis

The most recent of the 'Big Five' mass extinctions occurred 66 million years ago (Ma) at the Cretaceous-Palaeogene (K-Pg) boundary, primarily known for the demise of the non-avian dinosaurs and ammonites (Alvarez et al., 1980; Schulte et al., 2010; Brusatte et al., 2015). The effect of the K-Pg mass extinction on eutherian mammals (placentals and extinct relatives) is one of the most well-studied yet contentious inquiries of macroevolutionary biology. Eutherian mammals prior to the K-Pg boundary have been widely considered depauperate, small sized and limited in ecological variation, which contrasts the diversity seen in extant placentals. This traditional view, known as the *suppression hypothesis*, follows that the extinction of non-avian dinosaurs released eutherian mammals from constraints of predation and competition, stimulating a radiation at the onset of the Cenozoic (Osborn, 1902; Alroy, 1999; Codron et al., 2013; O'Leary et al., 2013). The most significant morphological evidence for the suppression hypothesis in the fossil record concerns a shift in body size disparity (Alroy, 1999). Mesozoic mammals generally weighed less than 100 g and rarely over 5 kg (Benevento et al., 2023). The maximum body mass of mammals increased exponentially and by roughly three orders of magnitude by the late Eocene (Cooper and Purvis, 2010; Smith et al., 2010; Raia et al., 2013; Slater, 2013). It is thought that this increase is indicative of a wider expansion of niche breadth as mammals filled the vacant niches left by large dinosaurs.

2.2 Early rise hypothesis

While the suppression hypothesis is strongly supported by studies that have investigated changes in body size (Alroy, 1999; Slater, 2013), body size is limited as a biodiversity metric for ecological variation since it can be influenced by several factors that may reflect stochastic rather than adaptive changes (Grossnickle, 2020). Recent studies instead focusing on morpho-functional traits (intrinsically linked to ecological function), such as jaw and dentition morphology, have instead suggested that eutherian diversity was attained through successive waves of diversification, starting far earlier than the K-Pg mass extinction (Grossnickle et al., 2019). The *early rise hypothesis* suggests an ecological diversification that occurred concurrently with the diversification of angiosperms and insects during and after the Cretaceous Terrestrial Revolution (125–80 Ma) (Figure 2) (Meredith et al., 2011; Benton et al., 2022). For example, using morphometric analyses of molar shape Grossnickle and Newham (2016) found that eutherian disparity began increasing during the Late Cretaceous. Wilson (2013) also found a similar pattern, as well as a decrease in molar disparity across K-Pg suggesting selective extinction. Ecological variation within Mesozoic mammals is not limited to diet; diversity in locomotor styles was reached as far back as the Jurassic in early eutherians (Chen et al., 2019).

2.3 Delayed rise hypothesis

Other studies have pointed towards the Paleocene-Eocene Thermal Maximum (PETM), an episode of elevated global temperatures 56 Ma that widened the latitudinal range of angiosperm-dominated forest, which in turn spurred the dispersal and radiation of eutherians (Figure 2) (Bininda-Emonds et al., 2007). Benevento et al. (2019) found that eutherian mammals had the most significant expansion in jaw morphospace at the onset of the Eocene. Bertrand et al. (2022) investigated endocranial volume in Cenozoic placentals and also found a delayed radiation in diversity, highlighting increases in body size during the Paleocene preceded an expansion in brain size in the Eocene, which became necessary as competition intensified in an increasingly mammal-dominated biota. While these alternative hypotheses do not necessarily refute the influence of the K-Pg mass extinction, they instead suggest the extinction accentuated evolutionary trajectories already underway.

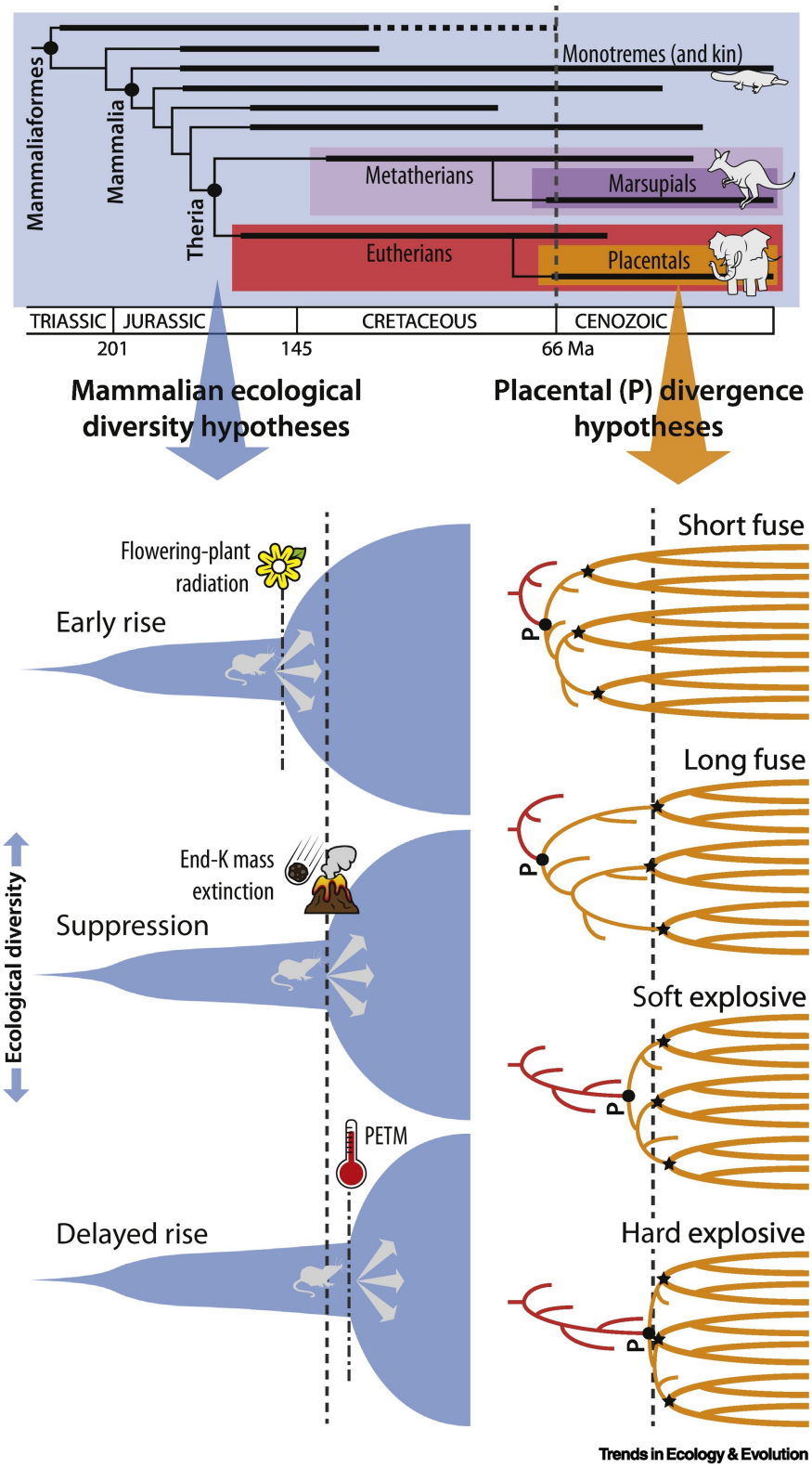

Figure 2: The contrasting ecological diversification triggers and placental origin hypotheses. Adapted from Grossnickle et al. (2019).

3 Disparity Analysis Methods

The conflicting findings of eutherian disparity across the K-Pg boundary raise the question: why do disparity analyses aiming to answer the same question arrive at different conclusions? Addressing this requires a critical look at the methodological underpinnings of disparity analyses.

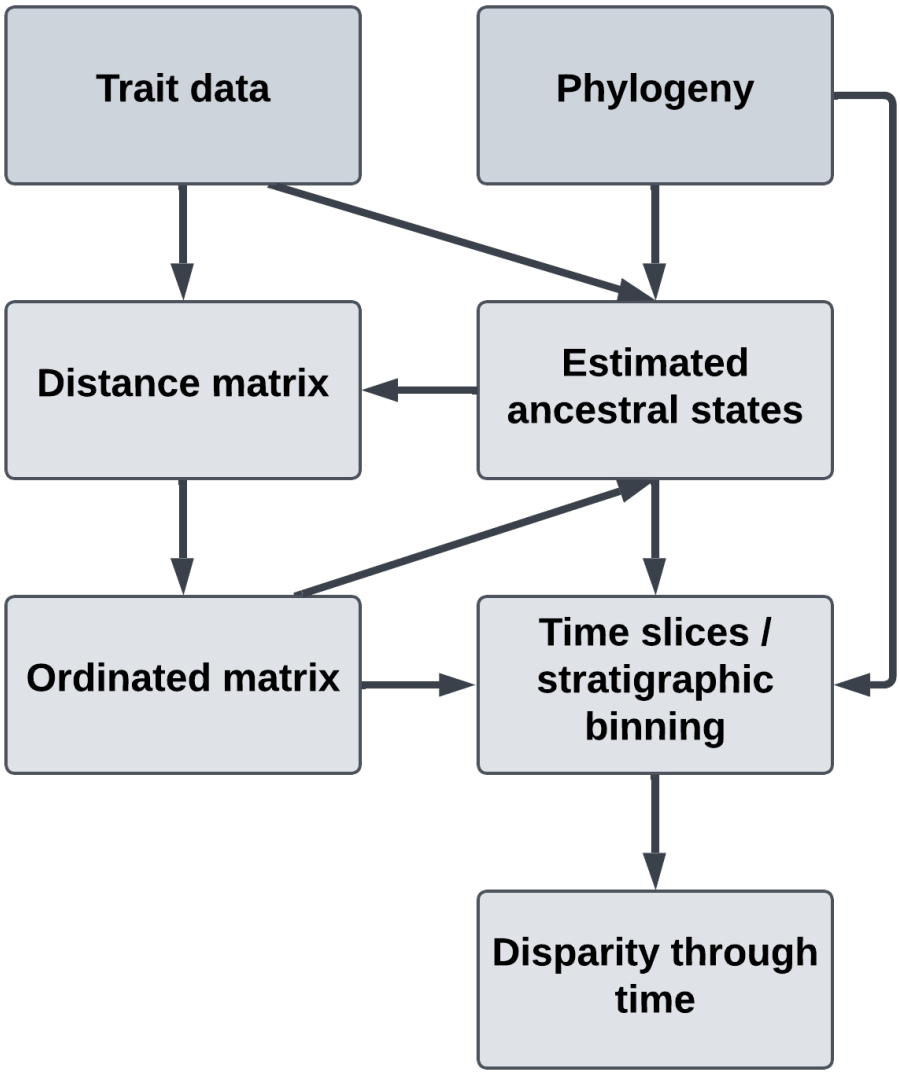

Figure 3: Disparity pipeline (N.B. this is specifically for discrete characters. Continuous characters would bypass the distance matrix after ASE or without ASE). As shown there are several routes to disparity, including bypassing ancestral state estimation (ASE), pre-ordination ASE, post-ordination ASE

For the purposes of this review, I focus on three key methodological choices: (1) phylogeny, (2) ancestral state estimation, (3) detecting shifts in disparity.

4 Phylogenetic hypotheses

The origin and tempo of lineage diversification in placental mammals has been a point of contention for decades, meaning there is no undisputed phylogenetic topology for eutherian mammals. Therefore, several studies investigating eutherian disparity have opted to time-bin the disparity data points by stratigraphy when assessing disparity through time (Grossnickle and Polly, 2013; Wilson, 2013; Grossnickle and Newham, 2016; Benevento et al., 2019). While this approach may be necessary in the absence of a well-resolved phylogeny, it inherently assumes a punctuated equilibrium model of morphological evolution. This assumption can be particularly problematic when assessing disparity across mass extinction events (Guillerme and Cooper, 2018). The instantaneous nature of the K-Pg extinction requires a framework capable of capturing morphological changes at fine temporal scales. Stratigraphic time binning, which aggregates data over broad intervals, risks obscuring critical, transient shifts in disparity that may occur within shorter durations following the extinction. Incorporating a phylogeny means a time-slicing method can be used (Guillerme and Cooper, 2018), which removes the punctuated assumption and can reveal more temporally refined patterns. There are currently three main phylogenetic hypotheses for placental mammals (Table 1):

Table 1: Summary of major models for the origin and diversification of placental mammals.

Model	**Placental origin**	**Evidence**	**References**
Hard explosive	66 Ma (K-Pg boundary)	Palaeontological: no crown placental fossils before K-Pg	Alroy (1999); O'Leary et al. (2013); Davies et al. (2017); Halliday et al. (2017)
Soft explosive	~ 80 Ma	Molecular: removed fossil calibrations for large (slow evolving) taxa	Phillips (2016); Phillips and Fruciano (2018)
Long fuse	$\sim 120 - 80$ Ma	Molecular: genome wide data	Archibald and Deutschman (2001); Meredith et al. (2011); dos Reis et al. (2012); Foley et al. (2016); Carlisle et al. (2023)

5 Ancestral state estimation

The poor resolution of the fossil record is an inevitable limitation affecting disparity analyses, since only a small biased portion of a clade's evolutionary history is preserved. The incorporation of a phylogeny together with trait data enables a switch from simply observing a pattern (points in morphospace) to inferring a process through time, through the use of ancestral state estimation (Lloyd, 2018).

5.1 Application of ancestral state estimation

Ancestral state estimation is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for tips on a phylogenetic tree (Schluter et al., 1997; Cunningham et al., 1998; Yang, 2006; Revell, 2025). Trait data may be continuous, such as body size, or discrete, such as diet type or presence/absence of a character. The application of ancestral state estimation has been widely utilised in phylogenetic comparative methods, including disparity.

Intuitively, ancestral state estimation is appealing for those studying deep-time patterns of disparity. Incorporating a phylogeny into disparity analyses will unavoidably introduce unsampled 'ghost lineages' (Brusatte et al., 2011). A ghost lineage is the hypothetical ancestor that is not observed in the fossil record, but can be inferred to have existed due to the presence of other fossils/extant taxa. This can be best explained as the branch on a phylogeny between two observed taxonomic units (Figure 4).

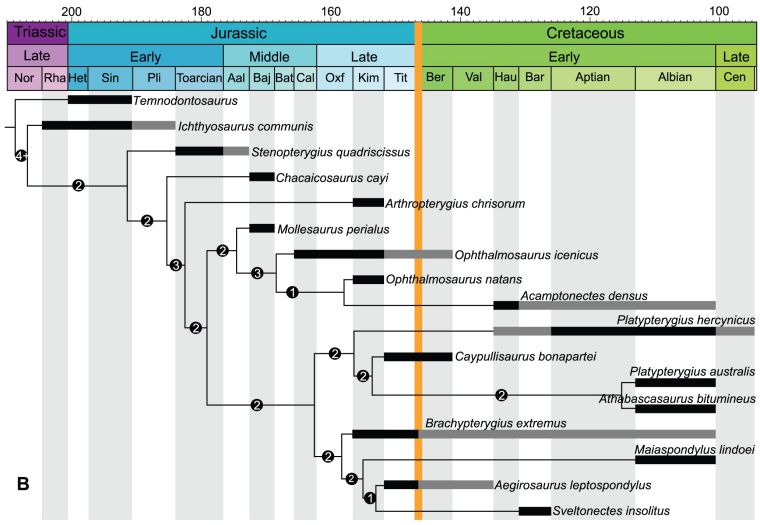

Figure 4: Phylogeny of ichthyosaurs showing observed stratigraphic range for each species (thick lines) and inferred ghost lineages (thin lines) (Fischer et al., 2012).

Ancestral state estimation has been employed as a way of inferring the traits of these hypothetical species based on the traits of their descendents. During ancestral state estimation, the estimated traits for nodes of a phylogeny are used as a way of sampling these ghost lineages. This fills in the gaps in the fossil record and alleviates the bias associated with uneven sampling across the phylogenetic tree (Friedman, 2009; Brusatte et al., 2011).

5.1.1 Likelihood methods

Likelihood methods for ancestral state estimation are based on a unifying statistical framework that treats the internal node states as parameters to be estimated, seeking to maximise the probability of the observed tip states given a topology and a model of evolution (Joy et al., 2016). The central tenet of the approach

is the likelihood function, which integrates information from the topology, which reflects the evolutionary history among taxa, and the model of trait evolution (which specifies how traits evolve along the branches). By systematically exploring the parameter space through numerical optimisation, the method search for the ancestral states that maximise the likelihood function. The same general likelihood framework underpins estimation of continuous and discrete characters; the core difference in estimation is the evolutionary model that is specified.

5.1.2 Continuous characters

Continuous trait evolution is typically modelled under a Brownian motion (BM) model of evolution (Felsenstein, 1985; O'Meara et al., 2006). BM models trait evolution as a directionless, stochastic process under which continuous characters undergo changes along the branches of a phylogeny. These evolutionary changes come from a Gaussian distribution with a mean of 0 and variance proportional to the product of the rate of evolution and time elapsed ($\sigma^2 t$). Therefore, the expected amount of trait divergence between any two taxa is proportional to the amount of shared evolutionary history according to the phylogeny.

Under BM, the vector ($\mathbf{x}$) of observed tip trait values follows a multivariate normal distribution (MVN) and has an expected value that is equal to the root state x_0.

$$\boldsymbol{x} \sim \mathrm{MVN}(x_0,\, \sigma^2 \boldsymbol{C})$$

$\boldsymbol{C}$ is a variance-covariance matrix for the shared ancestry of the taxa, defined by the tree height from the root to each tip (diagonal elements $\boldsymbol{C}_{ii}$) and the tree height above the root of the most recent common ancestor shared by each pair of tips (off-diagonal elements $\boldsymbol{C}_{ij}$).

When estimating ancestral states, the $\boldsymbol{C}$ matrix is expanded to include the shared history of the internal nodes. The evolutionary rate, σ^2, and root state x_0, are unknown parameters that can be estimated using Maximum Likelihood to give the multivariate normal distribution. The trait values at internal nodes are then calculated as the conditional expectations within this distribution. This means each node's estimated trait value is effectively a weighted average of the tip values, with the weights determined by the amount of shared evolutionary history (as defined by $\boldsymbol{C}$).

5.1.3 Discrete characters

Ancestral state estimation of discrete characters are modelled using a continuous-time Markov chain process, known as the Mk model (Pagel, 1994; Yang, 2006). This process is specified by a $k \times k$ matrix (k being the number of possible states), $\mathbf{Q}$, which gives the instantaneous transition rates (q) between states for each character.

$$\mathbf{Q} = \begin{bmatrix} -q_{0,1} & q_{0,1} \\ q_{1,0} & -q_{1,0} \end{bmatrix}$$

Markov chain transition matrices have the necessary properties that all off-diagonal entries are non-negative, all diagonal properties are non-positive and the sum of each row is zero. This is because in a continuous-time Markov process, probability flows out of a state (diagonal) and into other states (off-diagonal) and so the diagonal term represents the total rate of leaving a state.

The $\mathbf{Q}$ matrix is estimated using Maximum Likelihood; the value of $\mathbf{Q}$ that maximises the probability of the observed data given the tree topology. The matrix exponentiation of $\mathbf{Q}$ converts the instantaneous rates into probabilities of transitioning between states over a given time t:

$$\mathbf{P} = \exp(\mathbf{Q}t)$$

Global marginal estimation, which is most commonly implemented in software, fixes the value of $\mathbf{Q}$ and calculates the probability of each state at each node, given the observed data at the tips. These likelihoods are rescaled by their sum so that they add up to 1.0, and so represent an empirical Bayes posterior probability (Yang, 2006; Revell, 2025). Since this can become computationally intensive as the phylogeny increases in size, in practice functions use algorithms such as Felsenstein's pruning algorithm (Felsenstein, 1981).

5.2 Challenges in ancestral state estimation

"Character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem" (Swofford and Maddison, 1992).

5.2.1 Uncertainty in estimates

A critical issue within ancestral state estimation is that estimates tend to have broad uncertainty (Schluter et al., 1997). This uncertainty is not uniform across the tree: it typically increases towards root of the tree, where direct information from observed tip data becomes increasingly diffuse (Gascuel and Steel, 2014). For discrete characters, this uncertainty manifests as inconclusive scaled likelihoods, i.e., state 0 = 0.51, state 1 = 0.49. For continuous characters, uncertainty is wide confidence intervals.

How this uncertainty in estimations should be dealt with is equivocal. Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al., 2015), most computational implementations of ancestral state estimation return point estimates. For continuous characters, this will be the median value of the confidence interval and for discrete characters, it will be the most probable state according to a scaled likelihood threshold. By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses (such as down the disparity pipeline), all while obscuring potentially 'accurate' uncertainty (Ackerly et al., 2006; Vanderpoorten and Goffinet, 2006).

For discrete characters, there is a level of subjectivity in how to collapse the states into the most probable based on their scaled likelihood. It is common to have a threshold implementation in ancestral state estimation

software, whereby the user can set a scaled likelihood at which point states are collapsed to a point estimate, or left as uncertain, which is denoted as '0/1', for example. However, these thresholds are often arbitrarily set with no guidance of what is the 'correct' value (Smith and Donoghue, 2022; Casali et al., 2023). Much of the subjectivity relates to the trade-off between permissiveness and conservativeness; higher thresholds are likely too conservative and so too many states are left uncertain, perhaps missing true signals. Whereas lower thresholds may be too permissive and can lead to erroneous inferences. Interestingly, Keating (2023) highlights that uncertainty returned by a specified model is a better predictor of accuracy than likelihood diagnostics, such as AIC.

5.2.2 Sensitivity to fossil sampling

Given that uncertainty is often high in ancestral state estimation, how then can estimates be improved? Ané (2008) demonstrates that ancestral state estimation is *statistically inconsistent*, meaning that even if the sample size of extant tips is infinite, estimates are not expected to converge on the true value. Instead, the inclusion of fossils can improve estimates, due to the shorter evolutionary path to the root of the tree (Finarelli and Flynn, 2006; Finarelli and Goswami, 2013). Fossils can be used as priors on nodes, or simply added to the tree as tips.

By extension, inclusion of fossils can greatly influence inferences into disparity. Extant-only datasets often miss early rapid morphological diversification signals (Harmon et al., 2010), whereas analyses incorporating fossil data often finds the reverse (Hughes et al., 2013) For example, Mitchell (2015) found that analyses of bird disparity were underestimated when using ancestral state estimation of extant-only datasets.

5.2.3 Model dependency

Another key challenge is the sensitivity to violations of model assumptions. For continuous characters, this is particularly pertinent when traits have evolved under non-BM (Revell, 2025). When trait evolution deviates from BM – due to directional trends, bounded evolution, or regime shifts – ancestral state estimation can return biased estimations of ancestral states.

Both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. Trends mean a moving average through time, which violates the assumption of zero-mean change under BM. Similarly, issues estimating the evolutionary rate may arise when the trait data evolved under an Ornstein-Uhlenbeck (OU) model, whereby trait data evolves under a stabilising selection (Royer-Carenzi and Didier, 2016). It is common for regime shifts to occur at mass extinction boundaries (Jablonski, 1986). For example, Slater (2013) found that mammal body size was relatively constrained (OU) up until the K-Pg mass extinction which then allowed a release of constraint and BM evolution. Such shifts in regime could affect accurate estimation of evolutionary rate and thus state estimates.

As discussed previously, fossil sampling is crucial for accurate estimation, especially when model assump-

tions are violated (Puttick, 2016). Fossils assist in estimating and constraining model parameters, as well as distinguishing different evolutionary processes (Goswami and Clavel, 2025). For example, when estimating states under an OU model, fossils are necessary for estimating the primary optimum. Similarly, fossils would be necessary to detect a trend, such as Cope's rule that body size increases through time (Alroy, 1998). A similar logic applies to discrete characters; Keating (2023) found that inclusion of fossils improved estimate accuracy, irrespective of model misspecification.

6 Statistical tests in disparity

Given a calculation of disparity, it is necessary for this value to be statistically tested in the context of the research question. A popular choice is to carry out a PERMANOVA, which can be used to assess whether two groups share the same centroid and dispersion in the morphospace (Moon and Stubbs, 2020; Apaldetti et al., 2021; Donati et al., 2025). Alternatively, studies may opt to investigate change in disparity, for example through the use of a t test (Patton et al., 2021; Haug et al., 2023; Wang and Zhou, 2023). Therefore, a t-test may be preferred if investigating a nuanced change in morphospace that PERMANOVA cannot capture, such as change in variance.

The way these tests are implemented depends on the research question, but questions tend to relate to differences between groups (between clades for example) or through time (across time bins). However, using a single 'average change' statistic to compare disparity through time has limitations, notably that it ignores directional trends. Discounting trends in evolution may implicitly assume a punctuated model of evolution, where abrupt phenotypic shifts are interpreted as adaptive but are actually a reflection of stochastic processes (Hunt et al., 2015; Pagel et al., 2022). This punctuated bias is amplified when time binning disparity across stratigraphic bins. Despite its popularity, this method is flawed in that it only allows interpretation of changes in disparity between bins, rather than within. By aggregating data over broad intervals, the lack of resolution risks obscuring critical, transient shifts in disparity that may occur within shorter durations (following an extinction for example). This issue can be alleviated through time-slicing, whereby intervals are not punctuated by time bins but sub-sampling at equidistant points across a phylogeny (Guillerme and Cooper, 2018).

There have been several methods already developed for identifying non-random shifts in disparity. A common approach is to implement a null model to see if the empirical data deviates (Freckleton and Harvey, 2006; Murrell, 2018). For example, the global rank envelope test implemented by Murrell (2018) identifies non-random shifts as when the empirical Disparity-Through-Time curve (see Harmon et al. (2003)) falls within the most extreme 5% ranked curves of a null Brownian model. Similarly, Smith and Donoghue (2022) made use of simulations of discrete trait data as a null model based on the zero-force evolutionary law (the tendency for diversity to increase through time) (McShea and Brandon, 2010), to identify points at which disparity is greater/less than expected. However, a framework is needed that allows a statistical test to be

applied over large numbers of time series, when inspecting individual time series becomes impractical.

Part III

Thesis plan

7 Aims and objectives

Knowledge gap

The overarching aim of this PhD project is to understand the long term effect of mass extinctions by developping new state-of-the-art disparity methods. To achieve this I will be revisiting the critical question whether the K-PG mass extinction had an effect on eutherian disparity while focusing on improving the methodology of the disparity pipeline.

Research questions

- How robust are disparity analyses to uncertainty in ancestral state estimations?
- How can trend be accounted for when identifying non-random shifts in disparity?
- How do both selective and non-selective extinction events differ in the ability to recover accurate ancestral state estimations?
- Which of the three ecological hypotheses (Figure 2) for the diversification of eutherian mammals is most strongly supported by disparity analyses?

Research objectives

- Using a total-evidence fossilised birth-death phylogenetic inference approach to estimate long fuse, soft explosive and hard explosive tree distributions.
- Assess the accuracy of ancestral states estimations accuracy on disparity analyses across: 1) a range of models of trait evolution, 2) fossil sampling levels, 3) mass extinctions with different intensities and selectivities, 4) and disparity metrics.
- Develop a statistical framework, built on the BACI approach in ecology, that can identify immediate and long term trend changes in disparity at a specified time point.
- Use these methodological advances to build a pipeline capable of handling multiple tree distributions and disparity parameters that outputs a distribution of 'effect values' relating to K-Pg and eutherian disparity.

8 Chapter One: Eutherian phylogenies

Chapter aim: To estimate the phylogeny of eutherian mammals, using various calibrations for the placental node in order to recover the three common placental origin hypotheses: long fuse, soft explosive and hard explosive (section 4). The rationale of this chapter is so that these three tree distributions can be used in chapter four (section 11).

Research questions:

- When did major lineages of eutherians diversify?
- When did major shifts in evolutionary rates occur in eutherians?

Research objectives:

- I will use both molecular and morphological data for a total-evidence approach, which involves concatenating both molecular and morphological alignments into a single supermatrix
- The molecular and morphological matrices have been previously collected by my supervisor in a previous project
- Bayesian analysis will use RevBayes.
- I will implement different calibration strategies at the placental node to generate three alternative models: a long fuse model (placental node $\sim$ 100 Ma), a soft explosive scenario (placental node $\sim$ 80 Ma), and a hard explosive scenario (placental node 66 Ma).
- I will specify a Fossilized Birth-Death model in RevBayes to incorporate fossil tip ages and estimate divergence times.
- I will assess convergence and summarize the posterior distributions to obtain time-calibrated phylogenetic trees that reflect each calibration scenario.
- I will use RevGadgets to summarise the trees and plot figures.

9 Chapter Two: What is the effect of ancestral state estimations on disparity?

Chapter aim: To investigate how different approaches for treating uncertainties in ancestral state estimations affect estimates of morphological disparity under varying levels of fossil sampling. To expand this framework to understand how random and selective mass extinction events can confound ancestral state and disparity estimation accuracy.

Research questions:

- Which method of treating uncertainties recovers the most accurate ancestral state estimations?
- How does estimation accuracy change across different fossil sampling levels?
- How does estimation accuracy of disparity differ to accuracy of ancestral state estimation?
- What are the effects of mass extinction events on ancestral state estimations?
- How do random vs selective extinction events differ in their effect on ancestral state estimations?

Research objectives:

- Develop a simulation procedure that generates a distribution of 'true' evolutionary histories (i.e., phylogeny and trait data), with both discrete and continuous trait data.
- Remove varying levels of fossil tips from these 'true' phylogenies to simulate different fossil sampling levels.
- Run ancestral state estimation on these pruned trees/matrices with three different ways of treating uncertainty in estimates.
- Compare the raw distance between true vs estimated matrices.
- Compare the difference in disparity values between true vs estimated matrices.
- Run this procedure, but incorporating a mass extinction event into the simulations. These mass extinction events will have varying intensities and selectivities.

10 Chapter Three: Measuring changes in trends in disparity through time

Chapter aim: To develop a framework that can distinguish between random and non-random shifts in disparity through time, accounting for trend.

Research questions:

- How can traditional statistical tests, such as changes in average (t tests), be misleading in inferences into temporal change in disparity?
- How to implement a BACI (Before After Control Intervention) approach into disparity through time? The BACI framework, as laid out from an ecological perspective in Wauchope et al. (2021), aims to assess how interventions (which can be loosely defined as major climatic or geological events in this context) have an effect on a variable across a time series, given an estimate of the counterfactual.
- How to generate valid simulated control data, i.e., null models, for different specific questions?

- How to deal with time lags?

Research objectives:

- Building on the BACI framework laid out by Wauchope et al. (2021) I will develop a method to quantify shifts in disparity across a mass extinction (or any other fixed point in time that is suspected of having an effect on disparity).
- This framework will consider the mass extinction event to be an Intervention.
- Piecewise regression will be fit to data pre-extinction (Before) and post-extinction (After)
- Simulated data will be used as the null model (Control). The model parameters for replicating the pre-extinction trajectory will be estimated and simulations built on those parameters.
- BACI contrasts, i.e., $(A_I - B_I) - (A_C - B_C)$ can be used as a quantitative measure of the effect of the Intervention (mass extinction).
- Implement a modular approach to the function that allows for time lags to be accounted for. This would require the function to be able to detect breakpoints within a certain timeframe of the Intervention.

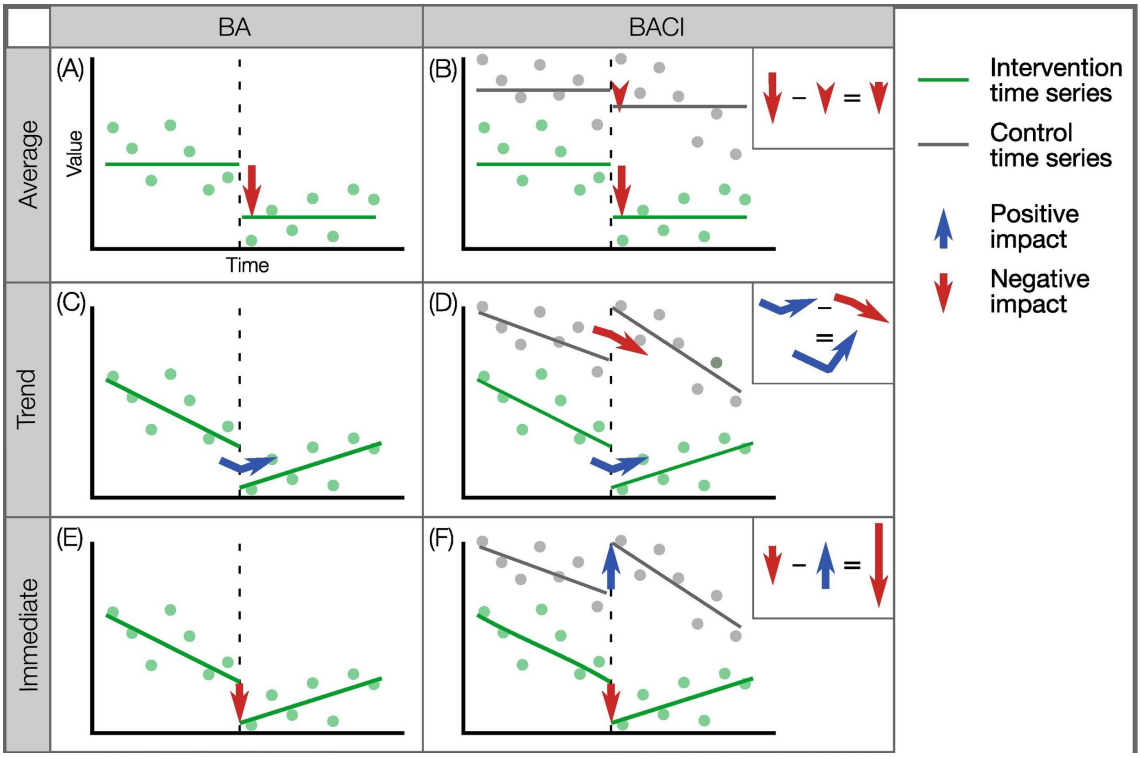

Figure 5: Blue arrows indicate positive change and red indicate negative change. The intervention effect can be interpreted either by change in average (A, B), change in trend (C, D) and/or an immediate change (E, F). BACI comparisons show the BACI contrast. For example, in panel D, there is a positive change in trend. Figure from Wauchope et al. (2021).

11 Chapter Four: Eutherian disparity over K-Pg mass extinction

Chapter aim: Incorporate the phylogenies from chapter one, the ancestral state method from chapter two and the test for non-random shifts in disparity from chapter two to develop a pipeline that can test whether the K-Pg mass extinction had an effect eutherian disparity.

Research questions:

- Is there a shift in trend of disparity at the K-Pg boundary?
- Does this shift depend on the phylogenetic topology used?
- Do patterns differ between different disparity metrics?

Research objectives:

- Use a dataset of discrete morphofunctional characters of jaw morphology and dentition, collected by Emmerson (2022).
- Develop a pipeline that iterates through different models and metrics: tree topology (long fuse/soft explosive/hard explosive), time slicing model and disparity metric.
- Use the BACI contrast function developed in chapter three to score a value of 'change in disparity' at 66 Ma to obtain a distribution of effects depending on the choice of topology, time-slicing model and disparity metric.

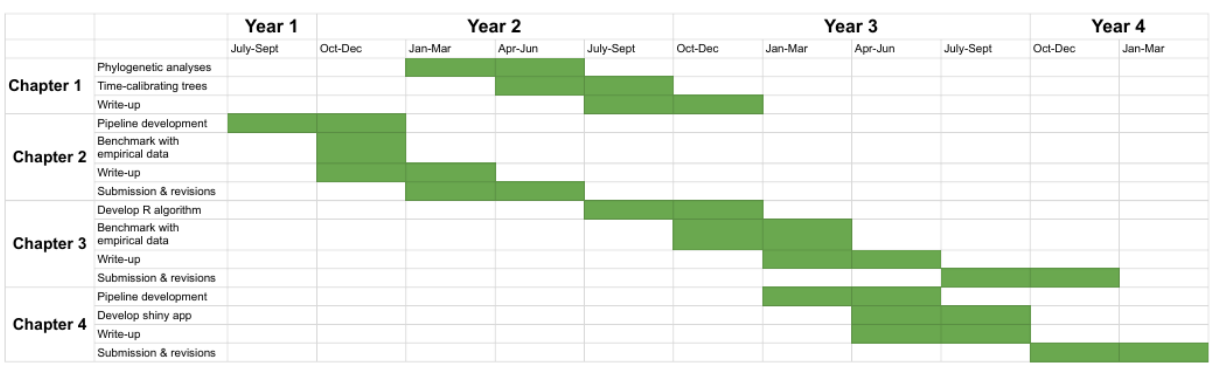

Figure 6: Gantt Chart for PhD timeline

Part IV

Summary of research achievements

Note that this is chapter two findings so far presented here.

12 Methods

12.1 Simulations

Simulations were carried out using `treats` (Guillerme, 2024). Trees were simulated using a modified version of the birth-death process algorithm. The process begins with a single lineage and iteratively selects lineages to grow the tree forward in time. At each step, all living branches are extended by a branch length increment, drawn from an exponential distribution based on the total number of lineages and the combined speciation and extinction rates. Selected lineages then undergo either speciation or extinction based on a probabilistic decision determined by the specified rates. 100 trees were simulated with birth-death parameters as speciation, $\lambda = 1.0$, and extinction, $\mu = 0.7$. Trees were fixed to have 50 extant tips and a range of 50 - 200 fossil tips.

Across each tree, discrete and continuous trait data were simulated. Discrete trait data were simulated using three different Markov chain matrices, representing fast (0.9), medium (0.1) and slow (0.01) transition rates, for 100 characters (85% binary and 15% three-state). Continuous trait data were simulated under Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models, with default parameters and for 50 characters. Trait data were simulated using the function `map.traits`, which assigns both nodes and tips trait values, essentially constructing full hypothetical evolutionary histories across each phylogeny for each trait type.

In order to simulate different levels of fossil sampling, each tree was split into four chronological 'stages' to replicate the stratigraphic nature of the fossil record. From each 'stage', fossil tips were randomly pruned from the tree and corresponding trait matrix at five levels:

- Living: only extant tips are retained
- Low: 5% fossils retained
- Medium: 15% fossils retained
- High: 50% fossils retained
- All: All fossils retained

I aim to modify this method in the future so that earlier stages have lower fossil sampling than more recent stages, to replicate the temporal gradient in completeness seen in the empirical fossil record (Davies et al., 2017).

12.2 Ancestral state estimation

Node states were then removed from the trait matrices leaving just tip states. Ancestral states were estimated using the `multi.ace` function in the `dispRity` package (Guillerme, 2018).

12.2.1 Discrete characters

For discrete state estimation, `multi.ace` is a wrapper for `castor::asr_mk_model`. Ancestral states were estimated using marginal likelihood estimation, whereby each state at each node is assigned a scaled likelihood (subsubsection 5.1.3). Symmetrical transition rates were set *a priori* during simulations and so the SYM model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating, 2023). Three different majority-rule approaches were implemented for collapsing the state probabilities into ancestral states. The *strict majority rule* uses a threshold that selects the state with a scaled likelihood equal or greater than a designated value (in this case 0.5). The *relative majority rule* sets a threshold at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states, k

$$\textit{Relative majority threshold} = \max(L_{\text{scaled}}) - \frac{1}{k}$$

The *no majority rule* (sampling approach) returns multiple matrices (x matrices) where ancestral states are probabilistically sampled based on their scaled likelihoods. For each node, the state assigned in each matrix is randomly drawn according to the probability distribution of the scaled likelihoods, meaning states with higher likelihoods are more likely to be selected across the sample of matrices. For example, a node with uncertain estimates (e.g., state 0 = 0.6, state 1 = 0.4) will be assigned state 0 in approximately 60% of the matrices and state 1 in 40% of the matrices. In this case, 100 matrices were returned using the no majority rule.

12.2.2 Continuous characters

Continuous ancestral states were estimated using the BM likelihood method described in subsubsection 5.1.2. Two approaches were applied to account for uncertainty in estimates. The *point estimate* approach simply returns the median of the 95% confidence interval. Alternatively, the *sampling* approach, whereby trait values are sampled across a normal distribution defined by the 95% confidence interval at each node, over a specified number of matrices (in this case 100).

12.3 Measuring raw distance

A Hamming distance metric to measure the raw accuracy at each node and state between the true and estimated discrete matrices. Both uncertain (where no state surpasses the threshold set by the majority rules) and inaccurate estimates were penalised with a score of 1, thus only correct estimates were not penalised (score of 0). To achieve fair comparison between each fossil sampling level, only the nodes present across all fossil sampling levels were included in these analyses (leaving 50 nodes). This outputted a distance matrix of dimensions N by C, whereby N is the number of nodes and C is the number of characters. The mean was taken from each matrix to output a distance score between 0–1.

For continuous characters, estimation error was quantified by calculating the Euclidean distance between

the estimated and true trait values at each node. Trait values were Z-score standardised prior to analysis to ensure fair comparison between BM and OU models.

12.4 Measuring error in disparity

To evaluate how ancestral state estimation influences disparity estimates, both the true and estimated trait matrices were analysed using a standard disparity pipeline (Figure 3). For discrete characters, the raw trait matrices were converted into distance matrices using the MORD metric (Lloyd, 2016), the most widely used distance metric in disparity analyses. These distance matrices were ordinated using principal coordinates analysis (PCO) to generate the multidimensional traitspaces. Disparity was quantified as the sum of variances across all PCO axes for each combination of tree, transition rate, fossil sampling level, and majority-rule method. To measure the estimation accuracy in disparity, the estimated disparity value was subtracted by the true disparity value and then divided by the true disparity value, to give a relative difference in disparity (this is necessary since true disparity values can vary greatly between different parameter combinations).

For continuous characters, the trait matrices were ordinated using principal component analysis (PCA) and then also disparity measured using the sum of variances across all PC axes. These analyses were performed using the `dispRity` package (Guillerme, 2018).

12.5 Extinction selectivity

These analyses will then be repeated, but incorporating mass extinction events into the trait and tree simulation process. A random mass extinction event can be applied to this

13 Results and Discussion

13.1 Discrete characters

Raw ancestral state estimation error across all nodes varies with the majority-rule method, transition rate and fossil sampling level (Figure 7). Across all majority rules, transition rates strongly influenced estimation accuracy. Estimation accuracy was consistently highest when transition rates were slow, with estimation accuracy decreasing as rates increased. The threshold-based strict and relative majority rule methods were less accurate in estimation compared to the sample-based approach for fast and medium transition rates, but attained equivalent accuracy for slow rates. Notably, fossil sampling level had a relatively negligible effect on accuracy, which contrasts previous findings (Finarelli and Flynn, 2006; Keating, 2023). Higher sampling slightly improved estimates under strict and relative majority rules at fast and medium rates. In contrast, the sample-based approach remained stable across all sampling levels, highlighting its robustness.

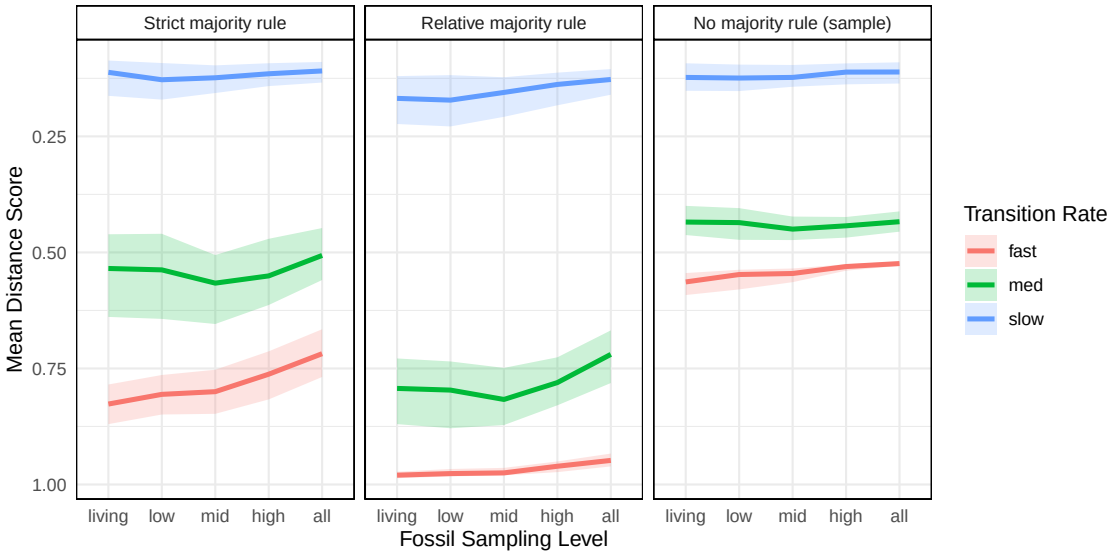

Figure 7: Mean distance of ancestral state estimations to true value across all nodes, plotted against fossil sampling level, for each majority rule and transition rate (fast = 0.9, medium = 0.1, slow = 0.01). The shaded area represents the upper and lower quartile range across the 100 tree replicates.

When assessing accuracy at the root node solely, similar patterns are observed, with the sample-based approach outperforming majority-rule methods (Figure 8). At the root node, fossil sampling has a greater impact, as increases in sampling increase estimation accuracy for medium and slow transition rates, but has little effect on fast rates. The importance of fossil inclusion is highlighted here, particularly at deeper nodes, when uncertainty increases due to the diffusion of informative observed data. However, the no majority rule (sampling) threshold method appears to be robust against low fossil sampling and high accuracy is maintained.

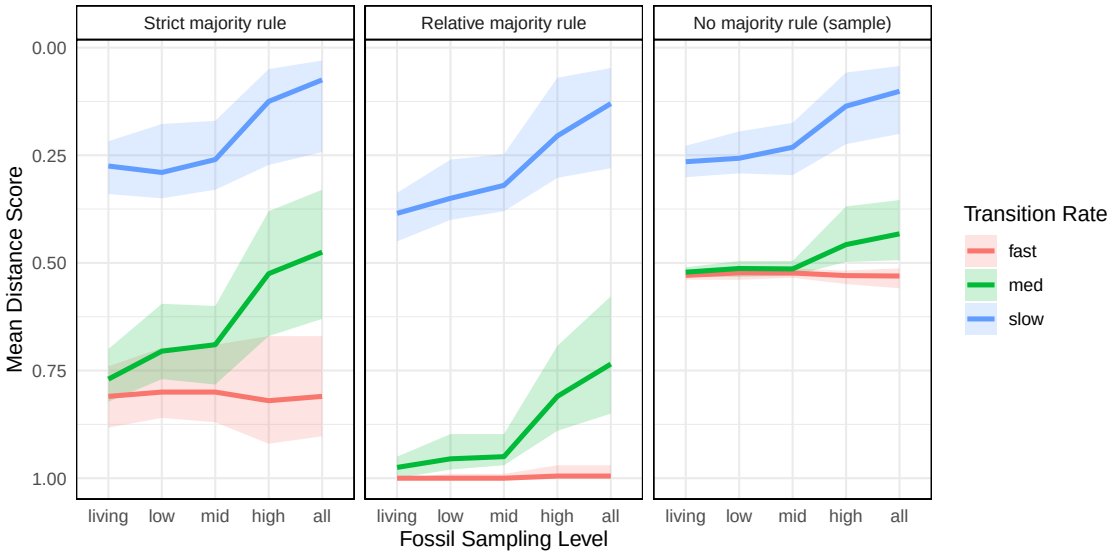

Figure 8: Mean estimation accuracy across at the root node for each tree.

Interestingly, the pattern observed when assessing the estimation of disparity differs to that of raw estimates (Figure 9). The sample-based approach again outperformed the majority-rule approaches, was robust against low fossil sampling and the true disparity value was recovered with high accuracy across all three transition rates. In contrast, the majority-rule approaches were more substantially affected by low fossil sampling, particularly at medium rates, which returned considerable variation in the disparity values returned. Somewhat counter-intuitively, the disparity value returned from fast transition rates were generally most accurate, despite having the lowest raw accuracy. This suggests that even when individual ancestral states are estimated poorly, the overall disparity signal can still be reliably captured

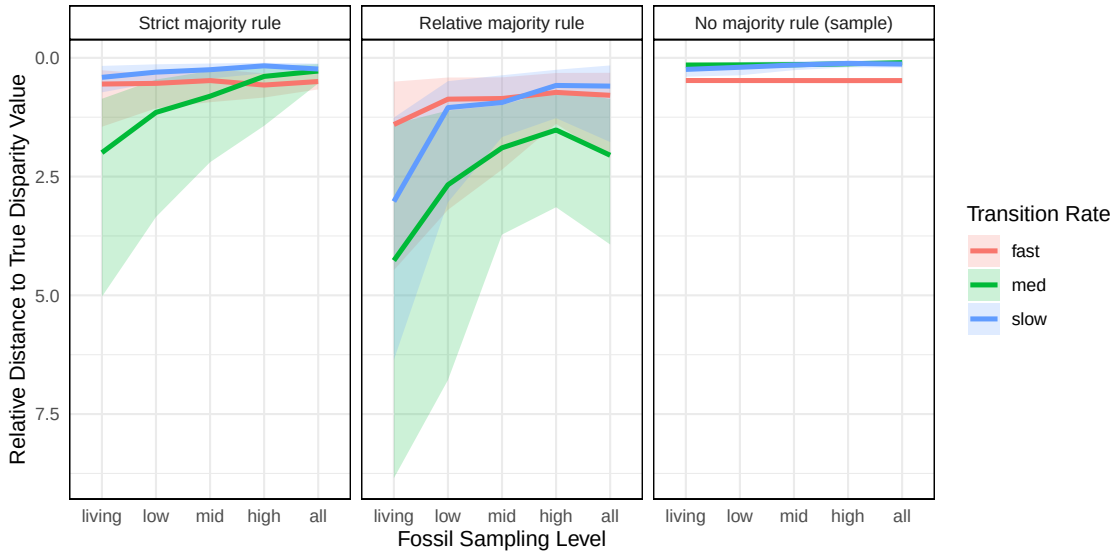

Figure 9: Mean estimation accuracy across at the root node for each tree.

From these preliminary findings I can suggest that, even when ancestral state estimation accuracy is poor at a 'raw state' level, accurate disparity signals are often still recovered. In addition, these findings highlight that accuracy should not be sacrificed for the sake of precision with ancestral state estimation. By incorporating uncertainty down the disparity pipeline using the sample-approach, as opposed to deterministic approaches such as the majority-rule, higher accuracy is recovered (Joy et al., 2016). For instance, when a true ancestral state has a posterior probability of 0.4, threshold-based approaches may discard it entirely, whereas the sample-based approach retains that state in 40% of reconstructions. This probabilistic representation enables more accurate recovery of trait variance and disparity, especially in cases where no single state dominates the posterior.

The incorporation of uncertainty in macroevolutionary workflows has been shown numerous times previously to yield more robust inferences (Slater and Pennell, 2014; Caetano and Harmon, 2019), yet ancestral state and disparity workflows often default to point estimates. These results highlight the utility of posterior sampling in ancestral state estimation and the robustness of disparity analyses to recovering true signals and thus should be the standard workflow for researchers employing these methods for discrete traits.

13.2 Continuous characters

Raw estimation accuracy for continuous characters was found to vary between models of trait evolution and method of handling uncertainty, but not across fossil sampling levels (Figure 10). Estimation accuracy of OU traits were more accurate than accuracy of BM trait estimation. This may seem intuitive, since estimation accuracy is expected to decrease as trait variance, σ, increases (Royer-Carenzi and Didier, 2016), although other studies have suggested that model misspecification (in this case OU when estimating traits under BM)

can lead to erroneous estimates (Ho and Ané, 2014). In an OU model, trait variance is restricted due to the stabilising selection, denoted by the α parameter. Therefore, it can be hypothesised from these results that the greater the strength of α, the more accurate the estimations (something that will be tested in future analyses). Furthermore, the fact that estimation accuracy does not improve across higher fossil sampling levels is surprising, particularly given that accurate estimation of the primary optimum/root state is thought to be only possible with fossils (Goswami and Clavel, 2025).

Interestingly, the sample-based approach returned less accurate estimates than using point estimates, contrasting the findings in discrete characters. This discrepancy may reflect the broader confidence intervals typically associated with continuous trait estimations. For discrete traits, estimations are constrained to finite number of possible states, so precision is inherently limited. In contrast, continuous traits exist on an infinite scale, making them more sensitive to imprecision, especially under models like BM where trait variance increases with time.

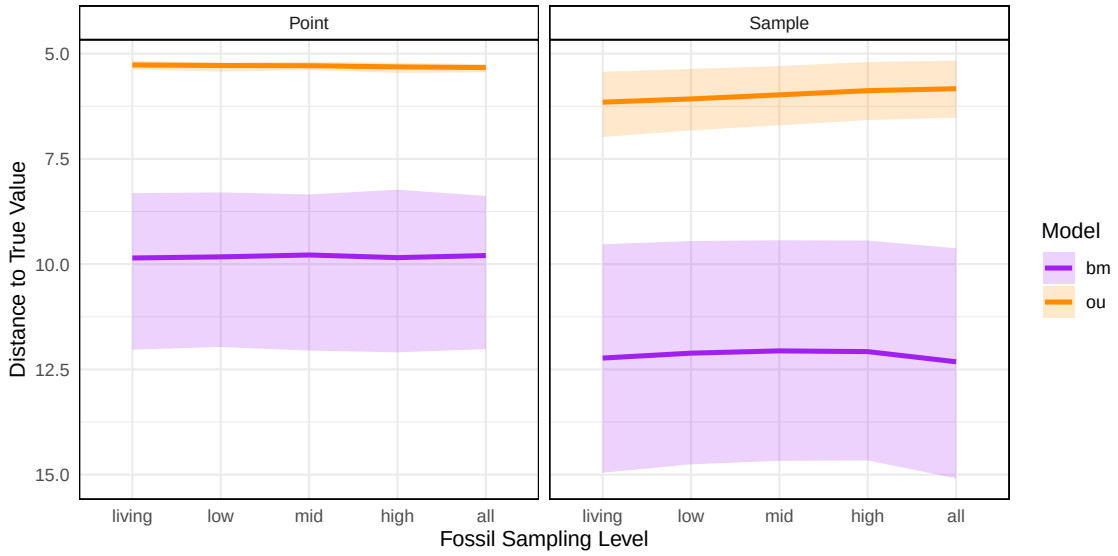

Figure 10: Mean estimation accuracy across at the root node for each tree.

Disparity estimation accuracy from continuous traits varied markedly between models and methods (Figure 11). Across all fossil sampling levels, estimates of BM traits derived from point estimates (left panel) were more accurate than those derived from a sample-approach. Estimates derived from sampling resulted in wide variation in estimates, which is likely due to the high uncertainty in the confidence intervals of estimated states. A contrasting pattern was observed for OU traits, with the sample-based approach recovering higher estimation accuracy than the point estimate approach, although both estimates had very low variance. Akin to the raw estimates of continuous characters, fossil sampling had little effect on estimation accuracy across both models.

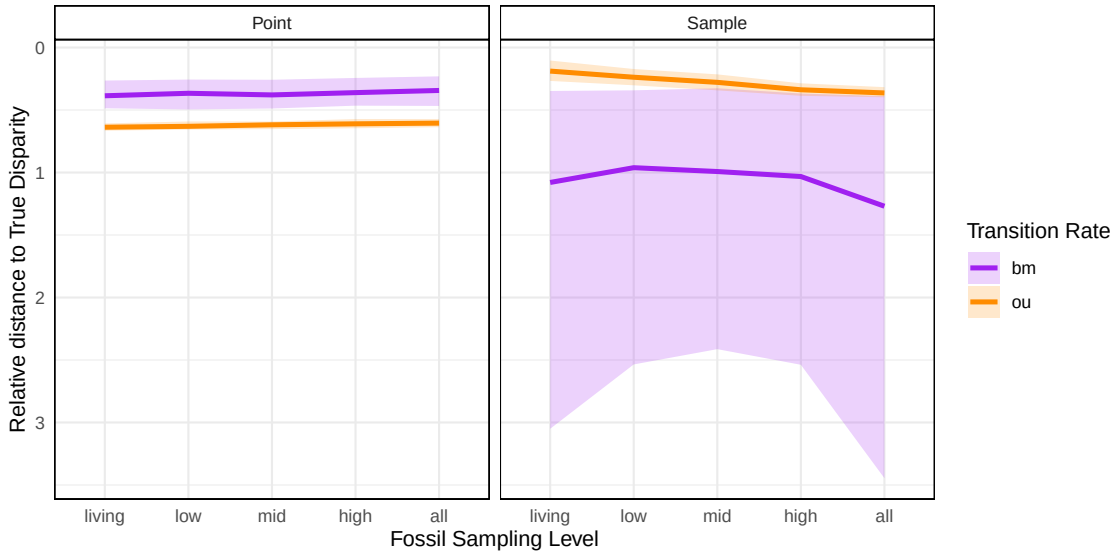

Figure 11: Mean estimation accuracy across at the root node for each tree.

14 Future directions

References

Ackerly, D., Schwilk, D. and Webb, C. (2006), 'Niche evolution and adaptive radiation: testing the order of trait divergence', *Ecology* **87**(sp7), S50–S61.

Ali, J. R., Blonder, B. W., Pigot, A. L. and Tobias, J. A. (2023), 'Bird extinctions threaten to cause disproportionate reductions of functional diversity and uniqueness', *Functional Ecology* **37**(1), 162–175.

Alroy, J. (1998), 'Cope's rule and the dynamics of body mass evolution in north american fossil mammals', *Science* **280**(5364), 731–734.

Alroy, J. (1999), 'The Fossil Record of North American Mammals: Evidence for a Paleocene Evolutionary Radiation', *Systematic Biology* **48**(1), 107–118. Publisher: Oxford Academic.

Alvarez, L. W., Alvarez, W., Asaro, F. and Michel, H. V. (1980), 'Extraterrestrial Cause for the Cretaceous-Tertiary Extinction', *Science* **208**(4448), 1095–1108. Publisher: American Association for the Advancement of Science.

Ané, C. (2008), 'Analysis of comparative data with hierarchical autocorrelation'.

Apaldetti, C., Pol, D., Ezcurra, M. D. and Martínez, R. N. (2021), 'Sauropodomorph evolution across the triassic–jurassic boundary: body size, locomotion, and their influence on morphological disparity', *Scientific Reports* **11**(1), 22534.

Archibald, J. D. and Deutschman, D. H. (2001), 'Quantitative Analysis of the Timing of the Origin and Diversification of Extant Placental Orders', *Journal of Mammalian Evolution* **8**(2), 107–124. Company: Springer Distributor: Springer Institution: Springer Label: Springer Number: 2 Publisher: Kluwer Academic Publishers-Plenum Publishers.

Bapst, D. W., Bullock, P. C., Melchin, M. J., Sheets, H. D. and Mitchell, C. E. (2012), 'Graptoloid diversity and disparity became decoupled during the Ordovician mass extinction', *Proceedings of the National Academy of Sciences* **109**(9), 3428–3433. Publisher: Proceedings of the National Academy of Sciences.

Benevento, G. L., Benson, R. B. J., Close, R. A. and Butler, R. J. (2023), 'Early Cenozoic increases in mammal diversity cannot be explained solely by expansion into larger body sizes', *Palaeontology* **66**(3), e12653.

Benevento, G. L., Benson, R. B. J. and Friedman, M. (2019), 'Patterns of mammalian jaw ecomorphological disparity during the Mesozoic/Cenozoic transition', *Proceedings of the Royal Society B: Biological Sciences* **286**(1902), 20190347.

Benton, M. J., Wilf, P. and Sauquet, H. (2022), 'The Angiosperm Terrestrial Revolution and the origins of modern biodiversity', *New Phytologist* **233**(5), 2017–2035. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/nph.17822.

Bertrand, O. C., Shelley, S. L., Williamson, T. E., Wible, J. R., Chester, S. G. B., Flynn, J. J., Holbrook, L. T., Lyson, T. R., Meng, J., Miller, I. M., Püschel, H. P., Smith, T., Spaulding, M., Tseng, Z. J. and Brusatte, S. L. (2022), 'Brawn before brains in placental mammals after the end-Cretaceous extinction', *Science* **376**(6588), 80–85. Publisher: American Association for the Advancement of Science.

Bininda-Emonds, O. R. P., Cardillo, M., Jones, K. E., MacPhee, R. D. E., Beck, R. M. D., Grenyer, R., Price, S. A., Vos, R. A., Gittleman, J. L. and Purvis, A. (2007), 'The delayed rise of present-day mammals', *Nature* **446**(7135), 507–512. Number: 7135 Publisher: Nature Publishing Group.

Brusatte, S. L., Butler, R. J., Barrett, P. M., Carrano, M. T., Evans, D. C., Lloyd, G. T., Mannion, P. D., Norell, M. A., Peppe, D. J., Upchurch, P. and Williamson, T. E. (2015), 'The extinction of the dinosaurs', *Biological Reviews* **90**(2), 628–642. Publisher: John Wiley & Sons, Ltd.

Brusatte, S. L., Montanari, S., Yi, H.-y. and Norell, M. A. (2011), 'Phylogenetic corrections for morphological disparity analysis: new methodology and case studies', *Paleobiology* **37**(1), 1–22.

Budd, G. E. (2021), 'Morphospace', *Current Biology* **31**(19), R1181–R1185.

Caetano, D. S. and Harmon, L. J. (2019), 'Estimating correlated rates of trait evolution with uncertainty', *Systematic Biology* **68**(3), 412–429.

Carlisle, E., Janis, C. M., Pisani, D., Donoghue, P. C. J. and Silvestro, D. (2023), 'A timescale for placental mammal diversification based on Bayesian modeling of the fossil record', *Current Biology* **33**(15), 3073–3082.e3. Publisher: Elsevier.

Casali, D. M., Boscaini, A., Gaudin, T. J. and Perini, F. A. (2023), 'Morphological disparity and evolutionary rates of cranial and postcranial characters in sloths (mammalia, pilosa, folivora)', *Palaeontology* **66**(1), e12639.

Chen, M., Strömberg, C. A. E. and Wilson, G. P. (2019), 'Assembly of modern mammal community structure driven by Late Cretaceous dental evolution, rise of flowering plants, and dinosaur demise', *Proceedings of the National Academy of Sciences* **116**(20), 9931–9940. Publisher: Proceedings of the National Academy of Sciences.

Codron, D., Carbone, C. and Clauss, M. (2013), 'Ecological Interactions in Dinosaur Communities: Influences of Small Offspring and Complex Ontogenetic Life Histories', *PLOS ONE* **8**(10), e77110. Publisher: Public Library of Science.

Cole, S. R. and Hopkins, M. J. (2021), 'Selectivity and the effect of mass extinctions on disparity and functional ecology', *Science Advances* **7**(19), eabf4072. Publisher: American Association for the Advancement of Science.

Cooper, N. and Purvis, A. (2010), 'Body Size Evolution in Mammals: Complexity in Tempo and Mode.', *The American Naturalist* **175**(6), 727–738. Publisher: The University of Chicago Press.

Cunningham, C. W., Omland, K. E. and Oakley, T. H. (1998), 'Reconstructing ancestral character states: a critical reappraisal', *Trends in ecology & evolution* **13**(9), 361–366.

Davies, T. W., Bell, M. A., Goswami, A. and Halliday, T. J. (2017), 'Completeness of the eutherian mammal fossil record and implications for reconstructing mammal evolution through the cretaceous/paleogene mass extinction', *Paleobiology* **43**(4), 521–536.

Donati, G. F. A., Albouy, C., Claverie, T., Mouillot, D., Govinden, R., Hagen, O., Ibrahim, S., Pagu, J., Zareer, I., Leprieur, F. et al. (2025), 'Continuity in morphological disparity in tropical reef fishes across evolutionary scales', *Communications Biology* **8**(1), 252.

dos Reis, M., Inoue, J., Hasegawa, M., Asher, R. J., Donoghue, P. C. J. and Yang, Z. (2012), 'Phylogenomic datasets provide both precision and accuracy in estimating the timescale of placental mammal phylogeny', *Proceedings of the Royal Society B: Biological Sciences* . Publisher: The Royal Society.

Edie, S. M., Jablonski, D. and Valentine, J. W. (2018), 'Contrasting responses of functional diversity to major losses in taxonomic diversity', *Proceedings of the National Academy of Sciences* **115**(4), 732–737. Publisher: Proceedings of the National Academy of Sciences.

Emmerson, R. M. (2022), Changes in morphological disparity in eutherian mammals across the K-Pg boundary and Palaeocene- Eocene Thermal Maximum using discrete morphofunctional characters, Master's thesis, University of Salford.

Felsenstein, J. (1981), 'Evolutionary trees from gene frequencies and quantitative characters: finding maximum likelihood estimates', *Evolution* pp. 1229–1242.

Felsenstein, J. (1985), 'Phylogenies and the comparative method', *The American Naturalist* **125**(1), 1–15.

Finarelli, J. A. and Flynn, J. J. (2006), 'Ancestral state reconstruction of body size in the caniformia (carnivora, mammalia): the effects of incorporating data from the fossil record', *Systematic Biology* **55**(2), 301–313.

Finarelli, J. A. and Goswami, A. (2013), 'Potential pitfalls of reconstructing deep time evolutionary history with only extant data, a case study using the canidae (mammalia, carnivora)', *Evolution* **67**(12), 3678–3685.

Fischer, V., Maisch, M. W., Naish, D., Kosma, R., Liston, J., Joger, U., Krüger, F. J., Pérez, J. P., Tainsh, J. and Appleby, R. M. (2012), 'New ophthalmosaurid ichthyosaurs from the european lower cretaceous demonstrate extensive ichthyosaur survival across the jurassic–cretaceous boundary', *PLoS One* **7**(1), e29234.

Foley, N. M., Springer, M. S. and Teeling, E. C. (2016), 'Mammal madness: is the mammal tree of life not yet resolved?', *Philosophical Transactions of the Royal Society B: Biological Sciences* . Publisher: The Royal Society.

Foote, M. (1993), 'Discordance and concordance between morphological and taxonomic diversity', *Paleobiology* **19**(2), 185–204.

Foote, M. (1994), 'Morphological disparity in ordovician-devonian crinoids and the early saturation of morphological space', *Paleobiology* **20**(3), 320–344.

Foote, M. (1997), 'The evolution of morphological diversity', *Annual Review of Ecology and Systematics* **28**(1), 129–152.

Freckleton, R. P. and Harvey, P. H. (2006), 'Detecting non-brownian trait evolution in adaptive radiations', *PLoS biology* **4**(11), e373.

Friedman, M. (2009), 'Ecomorphological selectivity among marine teleost fishes during the end-cretaceous extinction', *Proceedings of the National Academy of Sciences* **106**(13), 5218–5223.

Gascuel, O. and Steel, M. (2014), 'Predicting the ancestral character changes in a tree is typically easier than predicting the root state', *Systematic biology* **63**(3), 421–435.

Goswami, A. and Clavel, J. (2025), 'Morphological evolution in a time of phenomics', *Paleobiology* **51**(1), 195–213.

Grossnickle, D. M. (2020), 'Feeding ecology has a stronger evolutionary influence on functional morphology than on body mass in mammals', *Evolution* **74**(3), 610–628.

Grossnickle, D. M. and Newham, E. (2016), 'Therian mammals experience an ecomorphological radiation during the Late Cretaceous and selective extinction at the K–Pg boundary', *Proceedings of the Royal Society B: Biological Sciences* **283**(1832), 20160256.

Grossnickle, D. M. and Polly, P. D. (2013), 'Mammal disparity decreases during the Cretaceous angiosperm radiation', *Proceedings of the Royal Society B: Biological Sciences* **280**(1771), 20132110.

Grossnickle, D. M., Smith, S. M. and Wilson, G. P. (2019), 'Untangling the Multiple Ecological Radiations of Early Mammals', *Trends in Ecology & Evolution* **34**(10), 936–949.

Guillerme, T. (2018), 'disprity: a modular r package for measuring disparity', *Methods in Ecology and Evolution* **9**(7), 1755–1763.

Guillerme, T. (2024), 'treats: A modular r package for simulating trees and traits', *Methods in Ecology and Evolution* **15**(4), 647–656.

Guillerme, T. and Cooper, N. (2018), 'Time for a rethink: time sub-sampling methods in disparity-through-time analyses', *Palaeontology* **61**(4), 481–493.

Guillerme, T., Cooper, N., Brusatte, S. L., Davis, K. E., Jackson, A. L., Gerber, S., Goswami, A., Healy, K., Hopkins, M. J., Jones, M. E. et al. (2020), 'Disparities in the analysis of morphological disparity', *Biology letters* **16**(7), 20200199.

Guillerme, T., Puttick, M. N., Marcy, A. E. and Weisbecker, V. (2020), 'Shifting spaces: Which disparity or dissimilarity measurement best summarize occupancy in multidimensional spaces?', *Ecology and Evolution* **10**(14), 7261–7275. Publisher: John Wiley & Sons, Ltd.

Halliday, T. J. D., Upchurch, P. and Goswami, A. (2017), 'Resolving the relationships of Paleocene placental mammals', *Biological Reviews* **92**(1), 521–550. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/brv.12242.

Hansen, T. A., Kelley, P. H., Melland, V. D. and Graham, S. E. (1999), 'Effect of climate-related mass extinctions on escalation in molluscs', *Geology* **27**(12), 1139–1142.

Harmon, L. J., Losos, J. B., Jonathan Davies, T., Gillespie, R. G., Gittleman, J. L., Bryan Jennings, W., Kozak, K. H., McPeek, M. A., Moreno-Roark, F., Near, T. J. et al. (2010), 'Early bursts of body size and shape evolution are rare in comparative data', *Evolution* **64**(8), 2385–2396.

Harmon, L. J., Schulte, J. A., Larson, A. and Losos, J. B. (2003), 'Tempo and mode of evolutionary radiation in iguanian lizards', *Science* **301**(5635), 961–964.

Haug, C., Braig, F. and Haug, J. T. (2023), 'Quantitative analysis of lacewing larvae over more than 100 million years reveals a complex pattern of loss of morphological diversity', *Scientific Reports* **13**(1), 6127.

Ho, L. S. T. and Ané, C. (2014), 'Intrinsic inference difficulties for trait evolution with ornstein-uhlenbeck models', *Methods in Ecology and Evolution* **5**(11), 1133–1146.

Hopkins, M. (2013), 'Decoupling of taxonomic diversity and morphological disparity during decline of the cambrian trilobite family pterocephaliidae', *Journal of Evolutionary Biology* **26**(8), 1665–1676.

Hopkins, M. J. (2016), 'Magnitude versus direction of change and the contribution of macroevolutionary trends to morphological disparity', *Biological Journal of the Linnean Society* **118**(1), 116–130.

Hughes, M., Gerber, S. and Wills, M. A. (2013), 'Clades reach highest morphological disparity early in their evolution', *Proceedings of the National Academy of Sciences* **110**(34), 13875–13879.

Hull, P. (2015), 'Life in the Aftermath of Mass Extinctions', *Current Biology* **25**(19), R941–R952. Publisher: Elsevier.

Hunt, G., Hopkins, M. J. and Lidgard, S. (2015), 'Simple versus complex models of trait evolution and stasis as a response to environmental change', *Proceedings of the National Academy of Sciences* **112**(16), 4885–4890.

Jablonski, D. (1986), 'Background and mass extinctions: the alternation of macroevolutionary regimes', *Science* **231**(4734), 129–133.

Jablonski, D. (2001), 'Lessons from the past: Evolutionary impacts of mass extinctions', *Proceedings of the National Academy of Sciences* **98**(10), 5393–5398. Publisher: Proceedings of the National Academy of Sciences.

Jablonski, D. and Edie, S. M. (2025), 'Mass extinctions and their rebounds: a macroevolutionary framework', *Paleobiology* pp. 1–14.

Joy, J. B., Liang, R. H., McCloskey, R. M., Nguyen, T. and Poon, A. F. (2016), 'Ancestral reconstruction', *PLoS computational biology* **12**(7), e1004763.

Keating, J. N. (2023), 'What is the best method for estimating ancestral states from discrete characters?', *bioRxiv* pp. 2023–08.

Korn, D., Hopkins, M. J. and Walton, S. A. (2013), 'EXTINCTION SPACE-A METHOD FOR THE QUANTIFICATION AND CLASSIFICATION OF CHANGES IN MORPHOSPACE ACROSS EXTINCTION BOUNDARIES: EXTINCTION SPACE', *Evolution* pp. n/a–n/a.

Lloyd, G. T. (2016), 'Estimating morphological diversity and tempo with discrete character-taxon matrices: implementation, challenges, progress, and future directions', *Biological Journal of the Linnean Society* **118**(1), 131–151.

Lloyd, G. T. (2018), 'Journeys through discrete-character morphospace: synthesizing phylogeny, tempo, and disparity', *Palaeontology* **61**(5), 637–645.

Lowery, C. M. and Fraass, A. J. (2019), 'Morphospace expansion paces taxonomic diversification after end cretaceous mass extinction', *Nature ecology & evolution* **3**(6), 900–904.

Matthews, T. J., Triantis, K. A., Wayman, J. P., Martin, T. E., Hume, J. P., Cardoso, P., Faurby, S., Mendenhall, C. D., Dufour, P., Rigal, F. et al. (2024), 'The global loss of avian functional and phylogenetic diversity from anthropogenic extinctions', *Science* **386**(6717), 55–60.

McShea, D. W. and Brandon, R. N. (2010), *Biology's first law: the tendency for diversity and complexity to increase in evolutionary systems*, University of Chicago Press.

Meredith, R. W., Janečka, J. E., Gatesy, J., Ryder, O. A., Fisher, C. A., Teeling, E. C., Goodbla, A., Eizirik, E., Simão, T. L. L., Stadler, T., Rabosky, D. L., Honeycutt, R. L., Flynn, J. J., Ingram, C. M., Steiner, C., Williams, T. L., Robinson, T. J., Burk-Herrick, A., Westerman, M., Ayoub, N. A., Springer, M. S. and Murphy, W. J. (2011), 'Impacts of the Cretaceous Terrestrial Revolution and KPg Extinction on Mammal Diversification', *Science* **334**(6055), 521–524. Publisher: American Association for the Advancement of Science.

Mitchell, J. S. (2015), 'Extant-only comparative methods fail to recover the disparity preserved in the bird fossil record', *Evolution* **69**(9), 2414–2424.

Moon, B. C. and Stubbs, T. L. (2020), 'Early high rates and disparity in the evolution of ichthyosaurs', *Communications Biology* **3**(1), 68.

Murrell, D. J. (2018), 'A global envelope test to detect non-random bursts of trait evolution', *Methods in Ecology and Evolution* **9**(7), 1739–1748.

O'Leary, M. A., Bloch, J. I., Flynn, J. J., Gaudin, T. J., Giallombardo, A., Giannini, N. P., Goldberg, S. L., Kraatz, B. P., Luo, Z.-X., Meng, J., Ni, X., Novacek, M. J., Perini, F. A., Randall, Z. S., Rougier, G. W., Sargis, E. J., Silcox, M. T., Simmons, N. B., Spaulding, M., Velazco, P. M., Weksler, M., Wible, J. R. and Cirranello, A. L. (2013), 'The Placental Mammal Ancestor and the Post–K-Pg Radiation of Placentals', *Science* **339**(6120), 662–667. Publisher: American Association for the Advancement of Science.

O'Meara, B. C., Ané, C., Sanderson, M. J. and Wainwright, P. C. (2006), 'Testing for different rates of continuous trait evolution using likelihood', *Evolution* **60**(5), 922–933.

Osborn, H. F. (1902), 'The Law of Adaptive Radiation', *The American Naturalist* . Publisher: Ginn and Co.

Pagel, M. (1994), 'Detecting correlated evolution on phylogenies: a general method for the comparative analysis of discrete characters', *Proceedings of the Royal Society of London. Series B: Biological Sciences* **255**(1342), 37–45.

Pagel, M., O'Donovan, C. and Meade, A. (2022), 'General statistical model shows that macroevolutionary patterns and processes are consistent with darwinian gradualism', *Nature communications* **13**(1), 1113.

Patton, A. H., Harmon, L. J., del Rosario Castañeda, M., Frank, H. K., Donihue, C. M., Herrel, A. and Losos, J. B. (2021), 'When adaptive radiations collide: Different evolutionary trajectories between and within island and mainland lizard clades', *Proceedings of the National Academy of Sciences* **118**(42), e2024451118.

Phillips, M. J. (2016), 'Geomolecular Dating and the Origin of Placental Mammals', *Systematic Biology* **65**(3), 546–557. Publisher: Oxford Academic.

Phillips, M. J. and Fruciano, C. (2018), 'The soft explosive model of placental mammal evolution', *BMC Evolutionary Biology* **18**(1), 104.

Puttick, M. N. (2016), 'Partially incorrect fossil data augment analyses of discrete trait evolution in living species', *Biology Letters* **12**(8), 20160392.

Puttick, M. N., Guillerme, T. and Wills, M. A. (2020), 'The complex effects of mass extinctions on morphological disparity', *Evolution* **74**(10), 2207–2220.

Raia, P., Carotenuto, F., Passaro, F., Piras, P., Fulgione, D., Werdelin, L., Saarinen, J. and Fortelius, M. (2013), 'Rapid action in the Palaeogene, the relationship between phenotypic and taxonomic diversification in Coenozoic mammals', *Proceedings of the Royal Society B: Biological Sciences* **280**(1750), 20122244. Publisher: Royal Society.

Raup, D. M. (1981), 'Extintion: bad genes or bad luck?', *Acta geológica hispánica* pp. 25–33.

Raup, D. M. (1994), 'The role of extinction in evolution.', *Proceedings of the National Academy of Sciences* **91**(15), 6758–6763.

Raup, D. M. and Sepkoski, J. J. (1982), 'Mass Extinctions in the Marine Fossil Record', *Science* **215**(4539), 1501–1503. Publisher: American Association for the Advancement of Science.

Revell, L. J. (2025), 'Ancestral state reconstruction of phenotypic characters', *Evolutionary Biology* pp. 1–25.

Royer-Carenzi, M. and Didier, G. (2016), 'A comparison of ancestral state reconstruction methods for quantitative characters', *Journal of Theoretical Biology* **404**, 126–142.

Ruta, M., Angielczyk, K. D., Fröbisch, J. and Benton, M. J. (2013), 'Decoupling of morphological disparity and taxic diversity during the adaptive radiation of anomodont therapsids', *Proceedings of the Royal Society B: Biological Sciences* **280**(1768), 20131071.

Schluter, D. (2000), *The ecology of adaptive radiation*, OUP Oxford.

Schluter, D., Price, T., Mooers, A. Ø. and Ludwig, D. (1997), 'Likelihood of ancestor states in adaptive radiation', *Evolution* **51**(6), 1699–1711.

Schulte, P., Alegret, L., Arenillas, I., Arz, J. A., Barton, P. J., Bown, P. R., Bralower, T. J., Christeson, G. L., Claeys, P., Cockell, C. S., Collins, G. S., Deutsch, A., Goldin, T. J., Goto, K., Grajales-Nishimura, J. M., Grieve, R. A. F., Gulick, S. P. S., Johnson, K. R., Kiessling, W., Koeberl, C., Kring, D. A., MacLeod, K. G., Matsui, T., Melosh, J., Montanari, A., Morgan, J. V., Neal, C. R., Nichols, D. J., Norris, R. D., Pierazzo, E., Ravizza, G., Rebolledo-Vieyra, M., Reimold, W. U., Robin, E., Salge, T., Speijer, R. P., Sweet, A. R., Urrutia-Fucugauchi, J., Vajda, V., Whalen, M. T. and Willumsen, P. S. (2010), 'The Chicxulub Asteroid Impact and Mass Extinction at the Cretaceous-Paleogene Boundary', *Science* . Publisher: American Association for the Advancement of Science.

Sidlauskas, B. (2008), 'Continuous and arrested morphological diversification in sister clades of characiform fishes: a phylomorphospace approach', *Evolution* **62**(12), 3135–3156.

Silvestro, D., Kostikova, A., Litsios, G., Pearman, P. B. and Salamin, N. (2015), 'Measurement errors should always be incorporated in phylogenetic comparative analysis', *Methods in Ecology and Evolution* **6**(3), 340–346.

Slater, G. J. (2013), 'Phylogenetic evidence for a shift in the mode of mammalian body size evolution at the Cretaceous-Palaeogene boundary', *Methods in Ecology and Evolution* **4**(8), 734–744. Publisher: John Wiley & Sons, Ltd.

Slater, G. J. and Pennell, M. W. (2014), 'Robust regression and posterior predictive simulation increase power to detect early bursts of trait evolution', *Systematic Biology* **63**(3), 293–308.

Smith, F. A., Boyer, A. G., Brown, J. H., Costa, D. P., Dayan, T., Ernest, S. K. M., Evans, A. R., Fortelius, M., Gittleman, J. L., Hamilton, M. J., Harding, L. E., Lintulaakso, K., Lyons, S. K., McCain, C., Okie, J. G., Saarinen, J. J., Sibly, R. M., Stephens, P. R., Theodor, J. and Uhen, M. D. (2010), 'The Evolution of Maximum Body Size of Terrestrial Mammals', *Science* **330**(6008), 1216–1219. Publisher: American Association for the Advancement of Science.

Smith, T. J. and Donoghue, P. C. (2022), 'Evolution of fungal phenotypic disparity', *Nature Ecology & Evolution* **6**(10), 1489–1500.

Swofford, D. L. and Maddison, W. P. (1992), 'Parsimony, character-state reconstructions, and evolutionary inferences', *Systematics, historical ecology, and North American freshwater fishes* **1**.

Vanderpoorten, A. and Goffinet, B. (2006), 'Mapping uncertainty and phylogenetic uncertainty in ancestral character state reconstruction: an example in the moss genus brachytheciastrum', *Systematic Biology* **55**(6), 957–971.

Wan, J., Foster, W. J., Tian, L., Stubbs, T. L., Benton, M. J., Qiu, X. and Yuan, A. (2021), 'Decoupling of morphological disparity and taxonomic diversity during the end-permian mass extinction', *Paleobiology* **47**(3), 402–417.

Wang, M. and Zhou, Z. (2023), 'Low morphological disparity and decelerated rate of limb size evolution close to the origin of birds', *Nature Ecology & Evolution* **7**(8), 1257–1266.

Wauchope, H. S., Amano, T., Geldmann, J., Johnston, A., Simmons, B. I., Sutherland, W. J. and Jones, J. P. (2021), 'Evaluating impact using time-series data', *Trends in Ecology & Evolution* **36**(3), 196–205.

Webster, A. J. and Purvis, A. (2002), 'Testing the accuracy of methods for reconstructing ancestral states of continuous characters', *Proceedings of the Royal Society of London. Series B: Biological Sciences* **269**(1487), 143–149.

Wills, M. A., Briggs, D. E. G. and Fortey, R. A. (1994), 'Disparity as an evolutionary index: a comparison of Cambrian and Recent arthropods', *Paleobiology* **20**(2), 93–130.

Wilson, G. P. (2013), 'Mammals across the K/Pg boundary in northeastern Montana, U.S.A.: dental morphology and body-size patterns reveal extinction selectivity and immigrant-fueled ecospace filling', *Paleobiology* **39**(3), 429–469.

Yang, Z. (2006), *Computational molecular evolution*, OUP Oxford.